

Exercise:

- 1) Find the minimal polynomial of the matrix

$$A = \begin{bmatrix} 4 & 0 & 0 \\ 0 & 2 & 3 \\ 0 & 0 & 2 \end{bmatrix}$$

Since $A\vec{e}_1 = \begin{bmatrix} 4 \\ 0 \\ 0 \end{bmatrix} = 4\vec{e}_1$ we have that

$(A - 4I)\vec{e}_1 = \vec{0}$ and thus $\mu_1 = x - 4$.

Similarly as $A\vec{e}_2 = \begin{bmatrix} 0 \\ 2 \\ 0 \end{bmatrix} = 2\vec{e}_2$ we have that

$\mu_2 = x - 2$. Now as $A\vec{e}_3 = \begin{bmatrix} 0 \\ 3 \\ 2 \end{bmatrix}$ and

$A^2\vec{e}_3 = \begin{bmatrix} 4 & 0 & 0 \\ 0 & 2 & 3 \\ 0 & 0 & 2 \end{bmatrix} \begin{bmatrix} 0 \\ 3 \\ 2 \end{bmatrix} = \begin{bmatrix} 0 \\ 12 \\ 4 \end{bmatrix}$ we have that

$$A^2\vec{e}_3 = \boxed{4} A\vec{e}_3 + \boxed{-4} \vec{e}_3 \quad \text{or}$$

$(A^2 - 4A + 4I)\vec{e}_3 = \vec{0}$. Finally we conclude

that $\mu_3 = x^2 - 4x + 4 = (x - 2)^2$ and thus

$$\mu_A = \text{LCM}(x - 4, x - 2, (x - 2)^2) = (x - 2)^2 (x - 4).$$

* The power of 2 in μ_A indicates that A is not diagonalizable and that $\lambda=2$ is defective

Properties 15.1 Let $A \in \mathbb{C}^{n \times n}$ have characteristic polynomial of p_A and minimal polynomial μ_A . Then

- i. $\deg p_A = n$ and $\deg \mu_A \leq n$
- ii. $\mu_A \mid p_A$ as in $p_A = \mu_A \cdot q$ where q is also a polynomial
- iii. λ_i is an e-value of A if and only if $p_A(\lambda_i) = \mu_A(\lambda_i) = 0$.
- iv. If A has distinct e-values $\lambda_1, \dots, \lambda_n$ ($\lambda_i \neq \lambda_j$) then
$$\mu_A(x) = (-1)^n p_A(x) = \prod_{i=1}^n (x - \lambda_i)$$

Exercise:

2) Show that $\det A$ is the product of the

e-values of A .

$$p = |A - \lambda I|$$

$$p = p(-\lambda)$$

Assume that A has e-values of $\lambda_1, \dots, \lambda_n$ (possibly with repetitions) Now as

$p_A(\lambda) = |A - \lambda I|$ uniquely factorizes as $\prod_{i=1}^n (\lambda_i - \lambda)$ over $\mathbb{C}$ we have that

$$|A| = p_A(0) = \prod_{i=1}^n \lambda_i \text{ as desired.}$$

Diagonalization

Def 15.2 Let $T \in \mathcal{L}(\mathbb{C}^n, \mathbb{C}^n)$ and $A = [T]_{\mathcal{B}}$.

We say that T and therefore A is diagonalizable if there exists basis $\mathcal{B}'$ such that $[T]_{\mathcal{B}'}$ is diagonal.

Letting $A = [T]_{\{\vec{e}_i\}}$ and assuming that for

$\mathcal{B}' = \{\vec{v}_1, \dots, \vec{v}_n\}$, $[T]_{\mathcal{B}'} = \begin{bmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{bmatrix}$. Then

$T\vec{v}_i = \lambda_i \vec{v}_i$ and thus

$$\begin{bmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{bmatrix} = [\vec{v}_1 \cdots \vec{v}_n]^{-1} A [\vec{v}_1 \cdots \vec{v}_n]$$

or $A = [\vec{v}_1 \cdots \vec{v}_n] \begin{bmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{bmatrix} [\vec{v}_1 \cdots \vec{v}_n]^{-1}$

Theorem 15.3 If $A \in \mathbb{C}^{n \times n}$ has n linearly ind. e-vectors, $\{\vec{v}_1, \dots, \vec{v}_n\}$ with corresponding e-values $\lambda_1, \dots, \lambda_n$, then $A = S \Lambda S^{-1}$ where

$$S = [\vec{v}_1 \cdots \vec{v}_n] \text{ and } \Lambda = \begin{bmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{bmatrix}$$

* This is known as the diagonalization of A .

Exercise:

3) Find the diagonalization of the matrix

$$A = \begin{bmatrix} 4 & 4 \\ -2 & -5 \end{bmatrix}$$

$$\begin{aligned} |A - \lambda I| &= \begin{vmatrix} 4-\lambda & 4 \\ -2 & -5-\lambda \end{vmatrix} = (4-\lambda)(-5-\lambda) + 8 \\ &= \lambda^2 + \lambda - 12 \\ &= (\lambda - 3)(\lambda + 4) \end{aligned}$$

$$\lambda = 3, -4$$

$$\lambda_1 = 3: A - 3I = \begin{bmatrix} 1 & 4 \\ -2 & -8 \end{bmatrix}, \vec{v}_1 = \begin{bmatrix} 4 \\ -1 \end{bmatrix}$$

$$\lambda_2 = -4: A + 4I = \begin{bmatrix} 8 & 4 \\ -2 & -1 \end{bmatrix}, \vec{v}_2 = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$$

$$\therefore A = \begin{bmatrix} 4 & 1 \\ -1 & -2 \end{bmatrix} \begin{bmatrix} 3 & \\ & -4 \end{bmatrix} \begin{bmatrix} 4 & 1 \\ -1 & -2 \end{bmatrix}^{-1}$$

Similar Matrices

Def 15.4 Two matrices A and B are said to be similar iff there exists invertible matrix M such that $A = MBM^{-1}$. I.e. A and B both represent T and M is the change of basis.

Properties 15.5 Let $A = MBM^{-1}$, then

i. $|A| = |B|$

$$|A| = \cancel{|M|} |B| \cancel{|M|}^{-1}$$

ii. If $A\vec{v} = \lambda\vec{v}$, then $B(M\vec{v}) = \lambda(M\vec{v})$

iii. A and B have the same charact. and minimal polynomials

Note that if A and B have the same characteristic polynomial, then A need not be similar to B .

Matrix similarity is an equivalence relation

$$A \sim B \text{ iff } A = MBM^{-1} \text{ (iff } \mu_A = \mu_B)$$

i. (Reflexive) $A \sim A$

ii. (Symmetric) If $A \sim B$ then $B \sim A$

iii. (Transitive) If $A \sim B$ and $B \sim C$, then $A \sim C$

and thus $\sim$ partitions $\mathbb{C}^{n \times n}$ into disjoint equivalence classes $\{[A_j] : j \in J\}$ where

$$[A_j] = \{A : A \sim A_j\}.$$

Exercise :

4) Find all 2×2 matrices over $\mathbb{R}$ satisfying $A^4 = I$ up to equivalence classes.

$x-1$:

I

$x+1$:

$-I$

We begin by finding all possible minimal polynomials. Since $A^4 - I$ is the zero matrix, $x^4 - 1$ annihilates A and thus $\mu_A \mid x^4 - 1$.

x^2-1 :
 $\begin{bmatrix} 1 & -1 \end{bmatrix}$
 x^2+1 :
 $\begin{bmatrix} -1 & 1 \end{bmatrix}$

Now as $x^4-1 = (x-1)(x+1)(x^2+1)$ and
 thus μ_A must either, $\{x-1, x+1, x^2+1, x^2-1\}$. Finally we conclude that A
 must be similar to one of $\left\{ \underset{\theta=0}{I}, \underset{\theta=\pi}{-I}, \right.$
 $\left. \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}, \underset{\theta=\frac{\pi}{2}}{\begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}} \right\}$ which represent
 rotations of $\theta = 0, \frac{\pi}{2}, \pi$ and a reflection.

Steady - State Matrix

Def 15.6 Given $A \in \mathbb{C}^{n \times n}$, define the
 steady-state matrix as $A^\infty := \lim_{k \rightarrow \infty} A^k$ if
 said limit exists.

$$(A^\infty)_{ij} = \lim_{k \rightarrow \infty} (A^k)_{ij}$$

Theorem 15.7 Assuming that A is diagonalizable
 (true in general)

we have that $\lim_{k \rightarrow \infty} A^k$ exists iff the e-values
 of A satisfy $|\lambda| < 1$ or $\lambda = 1$.

Proof: Setting $A = S \Lambda S^{-1}$ we have that

$$A^k = S \Lambda^k S^{-1} = S \begin{bmatrix} \lambda_1^k & & \\ & \ddots & \\ & & \lambda_n^k \end{bmatrix} S^{-1}$$

$$\text{and thus } \lim_{k \rightarrow \infty} A^k = S \left(\lim_{k \rightarrow \infty} \begin{bmatrix} \lambda_1^k & & \\ & \ddots & \\ & & \lambda_n^k \end{bmatrix} \right) S^{-1}$$

exists iff $\lim_{k \rightarrow \infty} \lambda_i^k$ exists.

Markov Matrix

(Left Stochastic matrix)

Def 15-8 $M \in \mathbb{R}^{n \times n}$ is a Markov matrix iff the components of each column are non negative and sum up to 1.

Theorem 15-9 Letting M be as above so that $M_{i,j} \geq 0$ and $\sum_{i=1}^n M_{i,j} = 1$, then the e-values of M satisfy $-1 < \lambda \leq 1$. (There exist an e-value of λ)

Lemma A and A^T have the same e-values
since $|A^T - \lambda I| = |(A - \lambda I)^T| = |A - \lambda I|$

Proof of theorem: let λ be an e-value of M
 so that λ is also an e-value of M^T and
 let $\vec{v} \neq \vec{0}$ be such that $M^T \vec{v} = \lambda \vec{v}$. Now
 for any $1 \leq j \leq n$ j column of M = j row of M^T

$$\lambda v_j = \sum_{i=1}^n \underbrace{M_{i,j}}_{\text{j column of } M = \text{j row of } M^T} v_i$$

Now let $|v_k| = \max_{1 \leq j \leq n} |v_j|$ so that

$$\begin{aligned} |\lambda| |v_k| &= \left| \sum_i M_{i,k} v_i \right| \\ &\leq \sum_i M_{i,k} |v_i| \\ &\leq \sum_i M_{i,k} |v_k| \\ &= |v_k| \sum_i \cancel{M_{i,k}}^1 \end{aligned}$$

and hence $|\lambda| \leq 1$. Also since

$$M^T \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix} \quad \text{we have that}$$

M and M^T both have an e-value of 1.

Exercise

5) Given $M = \begin{bmatrix} 0.8 & 0.5 \\ 0.2 & 0.5 \end{bmatrix}$ find its

diagonalization and compute the steady-state matrix M^∞

$$|M - \lambda I| = \begin{vmatrix} 0.8 - \lambda & 0.5 \\ 0.2 & 0.5 - \lambda \end{vmatrix}$$

$$= \lambda^2 - \frac{13}{10}\lambda + \frac{3}{10}$$

$$= (\lambda - 1)\left(\lambda - \frac{3}{10}\right)$$

$$\lambda = 1, \frac{3}{10}$$